

torch.quantized_max_pool2d#

https://pytorch.org/docs/stable/generated/torch.quantized_max_pool2d.html

Description:

Applies a 2D max pooling over an input quantized tensor composed of several input planes. input (Tensor) – quantized tensor kernel_size (list of int) – the size of the sliding window stride (list of int, optional) – the stride of the sliding window padding (list of int, optional) – padding to be added on both sides, must be ≥ 0 and $\leq \text{kernel_size} / 2$

Function Signature

```
torch.quantized_max_pool2d(input, kernel_size, stride=[], padding=0, dilation=1, ceil_mode=False) → Tensor#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> qx = torch.quantize_per_tensor(torch.rand(2, 2, 2, 2), 1.5,
3, torch.quint8)
>>> torch.quantized_max_pool2d(qx, [2,2])
tensor([[[[1.5000]],

          [[1.5000]]],

        [[[0.0000]],

          [[0.0000]]]], size=(2, 2, 1, 1), dtype=torch.quint8,
quantization_scheme=torch.per_tensor_affine, scale=1.5,
zero_point=3)
```

Detailed Documentation

Documentation

Applies a 2D max pooling over an input quantized tensor composed of several input planes.

input (Tensor) – quantized tensor

kernel_size (list of int) – the size of the sliding window

stride (list of int, optional) – the stride of the sliding window

padding (list of int, optional) – padding to be added on both sides, must be ≥ 0 and $\leq \text{kernel_size} / 2$

dilation (list of int, optional) – The stride between elements within a sliding window, must be > 0 . Default 1

ceil_mode (bool, optional) – If True, will use ceil instead of floor to compute the output shape. Defaults to False.

A quantized tensor with max_pool2d applied.